Read me first — sim re-entry (540, July 2026)

Last synced: 2026-07-28 — ρ assortativity; **score ≠ select** wording; file map for later Charles.
OPORD: `../3-Master_Plan/re_entry/model_OPORD.md`
Your checklist: `../3-Master_Plan/re_entry/CHARLES_CHECKLIST.md`
If you feel lost: read `../3-Master_Plan/re_entry/00_READ_ME_FIRST.md` (docs 01–03) before opening sim code.

What to open vs what to ignore

Use daily (re-entry)	Ignore for daily work
This file + `540_three_step_sim.ipynb`	`tier1_cell10_playground_run.py` (**LEGACY** 538 widget UI)
`scripts/hero_model_reset_bundle.py` (Pass A)	Archived notebooks under `sports/archive/`
`scripts/540_rho_ablation_bundle.py` (Pass B)	`539_alex_model.ipynb` (older Alex lab; not 540 path)
`tier1_pool_assignment.py`, `tier1_generative_eda.py`, `tier1_sim_config.py`	Extending CELL 10 widgets

Engines (`tier1_*`) are **shared libraries**. Bundles are **thin runners**. Playground = old UI around the same engines.

Revolution vs evolution
| Do | Don't | |—-|——-| | Import `tier1_pool_assignment.py`, `tier1_generative_eda.py`, `tier1_sim_config.py` | Extend archived notebooks (`538D`, `538`, `537`, `535`) | | Thin scripts / `540_*` notebook | Copy-paste 538D cells | | Answer Alex-sized questions (knockouts, ablations) | Rivanna sweeps, bin-for-bin hero match as v1 gate |
Archived notebooks in `sports/archive/` are **SCOUT-era lab reference** — read only, do not edit.

Charles's three-step pipeline

1. **Assign** — build rosters (ρ or sort-and-chop)
2. **Score** — **S_i = A_i − λ·L_C** (Alex v1: **(B−D)=−L_C in the score**; λ lives here)
3. **Select** — winner rule: top **K** by **S_i** (later: soft / stochastic draw)

Knockouts (separate experiments):

Pass	What toggles	Status
A	λ=0 vs congestion in **S_i** (winner rule fixed)	Done → `alex_side_by_side_v0/`
B	ρ low / ρ high / sort-and-chop; **fix score + select**	`540_rho_ablation_bundle.py`

Three-step assignment (step 1 only)

0. Draw latent abilities **A_i**
1. Draw team targets **T_j** (e.g. Uniform on [−0.5, 0.5])
2. Sequential soft match with roster caps:

$$\pi_{ij} \propto \exp\left(-\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2}\right) \times (n_j + k)^{\alpha}$$

Optional preferential attachment: `USE_PREFERENTIAL_ATTACHMENT = True` → **α > 0**; default **False**.

ρ	Meaning
ρ = 0	Uniform among teams with roster room (max **mixing**)
ρ = 1	Moderate assortativity (≡ legacy **τ = σ ≈ 0.65**)
ρ high	Sharper match to nearest **T_j**
sort-and-chop	Separate code path — 537-style global sort + equal slices; **max assortativity benchmark**, not ρ→∞

σ (`ASSIGNMENT_SIGMA`, default **0.65**) is a fixed scale — not the user-facing knob.
Legacy docs use **τ (temperature)** with **opposite** intuition (small τ = assortative). **540+ uses ρ only** in prose and exports.

Hero estimand lock

Field	Value
Outcome	`Y_draft`, mean draft rate per bin
X (hero)	`poolq_loo`
Bins	16 quantile
Winsor	(0.01, 0.99)
min_minutes	20
Seasons	2011–2021

Slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Active files

Role	Path
Empirical	`530_sports_pipeline.ipynb`, `sports_pipeline/`
Sim engines	`tier1_pool_assignment.py`, `tier1_generative_eda.py`, `tier1_sim_config.py`
Pass A bundle	`scripts/hero_model_reset_bundle.py` → `alex_side_by_side_v0/`
Pass B bundle	`scripts/540_rho_ablation_bundle.py` → `alex_rho_ablation_v0/`
Notebook	`540_three_step_sim.ipynb`
Binding	`3-Master_Plan/BINDING_Selection_is_its_own_step.md`
Slide	`3-Master_Plan/re_entry/Model.pdf`

Export naming

```
sports/datasets/mbb/exports_inverted_u_v0/
  alex_side_by_side_v0/          # Pass A (λ knockout)
  alex_rho_ablation_v0/          # Pass B (ρ ablation)
```

Pass B artifacts: `generative_rho_low_16quantile.csv`, `generative_rho_high_16quantile.csv`, `generative_sort_chop_16quantile.csv`, `rho_ablation_summary.txt`, `rho_ablation_caption.txt`, PNG.

Anti-patterns

- Opening `archive/` notebooks as the daily workspace
- Merging **L_net** and **S_i** in comments or claims
- Saying ρ ablation **proves** hero bin-for-bin match
- Using τ in new user-facing text (archive reference only)

When stuck

Question	Read
What are we proving?	`re_entry/02`, `Model.pdf`
Assignment math (legacy τ)	`documents/Tier1_Presorting_Design_Note.md`
Widget-era manual	`documents/538_Cell10_Generative_Manual.md` (archive context)
Implementation	`tier1_pool_assignment.py` — `soft_assign`, `simulate_generative_rosters`